

Intonational variation in Swiss German

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The present study offers preliminary results on the intonation of Swiss German dialects spoken in Berne, Brig, and Zurich. In following a predominantly phonetic approach, the comparison of the subjects from each location allows for a distinction between region-specific and individual intonational characteristics; while the comparison between the different recording locations offers insight into the geolinguistic structure of intonation. Following the extraction of time-related parameters as well as the fundamental frequencies, intonation models for each subject as well as for each recording location are created. The primary goal is to analyse the nature and the extent of global as well as local intonation units present in the subjects at hand. This work delves into a largely untouched area of Swiss German geolinguistics and sheds light on the state of intonational variation in Swiss German and also discusses some of the possible reasons for variation in intonation in the examined dialects.

1. Introduction

Dialectology has described many features of Swiss German in great detail. Particularly segmental phonetics and phonology, as well as morphology, have been researched thoroughly. The Linguistic Atlas of German Switzerland (1962), a six-volume work published between 1962 and 1997, constitutes the basic bulk of research on Swiss German dialects, aside from the *Idiotikon* (1881 ff.), the wordbook of Swiss German. Within morphology and the lexicon there also exist numerous dialect-specific wordbooks and grammars. Yet, these dialect grammars and the Linguistic Atlas of Switzerland do not consider syntactic features. This research gap is being filled by a project at the University of Zurich, where research is currently taking place on a Syntax Atlas of German Switzerland (cf. Seiler 2005). It is conspicuous, however, that very little research has been conducted on prosodic features of the dialects. This is where our current SNSF project “Quantitative Approaches to a Geolinguistics of Swiss German Prosody” pitches in. Firstly, this

paper gives a brief sketch of what prosodic research entails, before an overview of some of the most relevant literature on Swiss German prosody is given. Then, our methodology is described, followed by a short introduction of the intonation model that we apply in our project, the Fujisaki model, before we delve into the preliminary results of the intonation of 15 speakers.

2. Swiss German prosody

The word prosody originally comes from Greek *prosoidia*, which refers to that which is sung onto speech. The prosodic form thus consists primarily of the melodic form of an utterance, yet it is also manifested in the loudness and durational features. The acoustic correlate of pitch is F0, while the correlate of loudness is the intensity of an utterance. The durational features of an utterance are described by the timing and the phrasing. By phrasing we mean the setting of phrase boundaries within an utterance. We understand timing as the duration of segments. The present paper looks at the melodic, i.e., intonational, characteristics of Swiss German dialects.

With regard to Swiss German, the earliest grammar writings that also address the intonational features were at the beginning of the 20th century. These works, created between 1910 and 1941, are collected in the *Beiträgen zur schweizer-deutschen Grammatik*, which, among many other contributions, include Elisa Wipf's "Mundart von Visperterminen im Wallis" (1910). By means of impressionistic terminology she describes the Wallis dialect as a singing dialect. Apart from such merely perceptual descriptions, systematic descriptions generally appeared with the rise of experimental phonetics (see below). The recording and analysis of the speech signal triggered an interest in intonation which consequently boosted research in that area.

Apart from these early accounts of intonation, there is only a four page conference paper by Fitzpatrick (1999), and the results of our earlier study: there are no other modern descriptions of Swiss German prosody. From the language typology perspective, Fitzpatrick investigates the intonation of declaratives – where she contrasts Northern Standard German and Bernese German. She concludes that in Bernese German, GToBI labels of the kind L*+H are more common, while in Northern Standard German H*+L is more prevalent. The previous study, titled "Creating the foundations for an investigation of Swiss German prosody" (Häsler et al. 2005), looked at the intonation of 3 subjects – 2 from Berne and 1 speaker from Zurich. As for the methodology, very similar strategies as in the present project were used. In terms of results we find overlap between the two projects, which will be looked at in the subsequent section.

3. Methods

In our project we study spontaneous speech of Bernese and Valais Swiss German on the suprasegmental level (Zurich and Chur will follow). In an interview setting, 41 Gymnasium (High School) students were asked questions largely on the topic of the subjects' plans after school. For the study at hand, 15 speakers were analyzed; 8 from the Midland dialect of Berne (henceforth BE) and 7 from Brig, representing the Alpine, Valais (henceforth WS) dialect. The sound files were then labelled on the segmental level in PRAAT (2007) and annotated with factors that are known to influence the fundamental frequency of these segments, such as word class, syllable nuclei, phrase boundaries etc. The database consists of roughly 1500 syllables per speaker. For intonation, pitch values were extracted and pitch contours were subsequently smoothed. For the latter, microprosodic F0 perturbations at the segmental level which affect the shape of the contour but do not significantly affect the listener's perception of tone height were deleted. Generally, these are due to the effects of stops on the neighbouring vowels or F0 dips in voiced fricatives.

The Fujisaki (1983) model is used for the analysis. It describes intonation as the superposition of a phrase component (henceforth Ap), representing the long-term F0 alterations over the domain of an intonational phrase (IP), and of an accent component (henceforth Aa), often word accents, which can, however, stretch over several syllables. We opted for this approach because the model fairly accurately generates and reproduces virtually any given F0 contour through a mathematical decomposition of the contour. This parametrisation is beneficial in that it enables quantification and modelling of the F0 representations. Two drawbacks of the model are possibly on the one hand its difficulty with modelling steep local F0 falls – which cannot be modelled optimally. Furthermore, slow-rises – as far as the original Fujisaki model is concerned – must be modelled either as a sequence of local accent commands that increase in amplitude towards the end of the phrase or, alternatively, by placing a number of phrase commands in short intervals. Both of these approaches do not lead to a sensible linguistic interpretation of the parameters – thus we are currently working on an integration of an additional, a slow-rise component (introduced by Mixdorff in 1998, see references) which allows the modelling of slow-rises more accurately.

The Fujisaki parameters were extracted with a tool provided by Mixdorff (2008). These parameters are then linked to the linguistic parameters, i.e., the ones that the researcher attributes to each accent or phrase command to the corresponding syllables, based largely on the auditory analysis and, to a lesser extent, on the visual inspection of the F0 contour. The results of the analysis can subsequently be analysed statistically. This is where we correlate the extracted parameters with parameters from the annotation step. The statistical analyses are performed at several levels.

4. Results

When listening to a person speak in a WS dialect and a person with a BE dialect, one can clearly perceive melodic differences. The question we ask ourselves is what intonational features are different from one dialect to another. Apart from pointing out that the WS dialect is perceived as a singing variety, Wipf (1910) mentions that they place pitch accents not only on stressed, but also on unstressed and schwa syllables. From this it follows that they should generally demonstrate more local accents than the BE variety. We therefore tested the following three hypotheses:

1. WS speakers generally produce more Aas than the BE group;
2. WS speakers put more Aas on grammatical words, on unstressed syllables, and on schwa syllables than the BE;
3. WS speakers' pitch range is higher than that of the BE, i.e., their Aa and Ap amplitudes are assumed to be more distinct.

4.1 Results I: Number of Aas produced

As early as 1910, Wipf claimed that WS speakers generally place more word accents than other Swiss German speakers do. As for the raw number of produced Aas, it turns out that the WS show 112 Aas/min, while the BE only produce 94 Aas/min. This indicates that on the absolute level the WS in fact realise more local accents than the BE do (t-test, $p=0.035$) – which, among other factors, may be one of the reasons they are perceived as a singing variety of Swiss German. However, it must be borne in mind that the articulation rate, too, has an effect on the number of accent commands produced. If someone is a fast speaker, he/she can, naturally, produce more intonational phrases and accents. One measure for the articulation rate is the number of syllables produced per second. Mean values suggest that the WS on average realise 5.8 syllables/second, while the BE only speak at the rate of 4.9 syllables/second. In order to correctly compare the number of local accents produced, then, articulation rate, too, must be taken into account. If the BE group's articulation rate is levelled with that of the WS, then, on a relative level, the WS and BE produce almost the exact same number of Aas, namely 110 Aas/min for the BE group and 112 Aas/min for the WS group; which is to say that our perception of the WS group's higher number of Aas/min is clearly related to their articulation rate.

4.2 Results II: Word class, unstressed syllables, and schwa syllables

In both observed Swiss German dialects we find that the vast majority of local accents fall on lexical words. The WS place 74% of all Aas on lexical words, the

BE only 68% (Chi Square 0.0001). This result indicates that the investigated WS speakers stress lexical words more rigorously with fast, local, F0 movements than the BE speakers do.

When we look at word stress as a parameter that affects intonation, we, somewhat surprisingly, find that both groups produce approximately 56% of all Aas on stressed syllables, 38% on unstressed syllables, and 6% on schwa syllables. This result is rather unexpected, as there are no significant differences between the BE and the WS groups. It is thus not, as we had speculated in one of our hypotheses, a difference in the number of Aas put on unstressed syllables and schwa syllables that can account for the melodic differences between the dialects; rather, these differences can possibly be further found in the last parameter of investigation: pitch range.

4.3 Results III: Aa and Ap amplitudes

Aa and Ap amplitude denote the extent to which a person varies his/her pitch, i.e., the pitch range. When we look at the phrase command amplitude we find considerable variation between the two speaker groups. Figure 1 shows Ap amplitudes for each speaker.

It turns out that the WS speakers show higher Ap amplitudes than the BE speakers (t-test, $p=0.0022$), which implies that in terms of F0 variation at the global level the WS vary more extensively. Yet, it must be added that the phrase commands of both

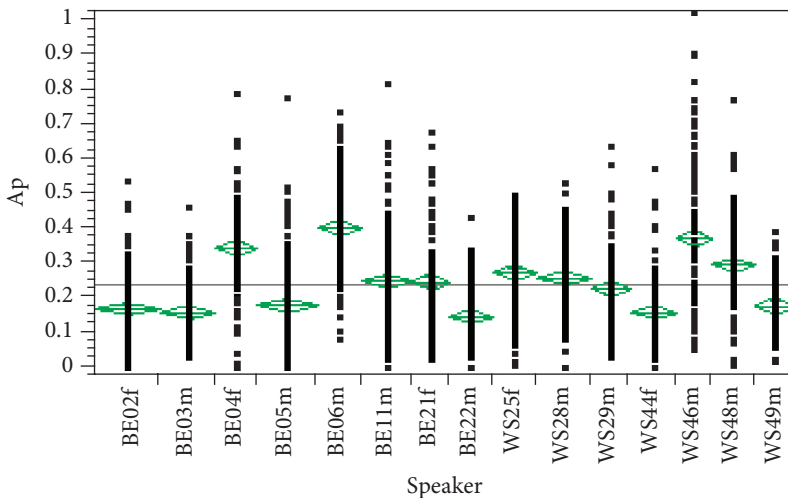


Figure 1. Ap amplitudes. Y-axis: Amplitude, X-axis: speakers. Diamonds mark speaker means and the 95%-confidence interval. The overall mean is marked with the grey horizontal line.

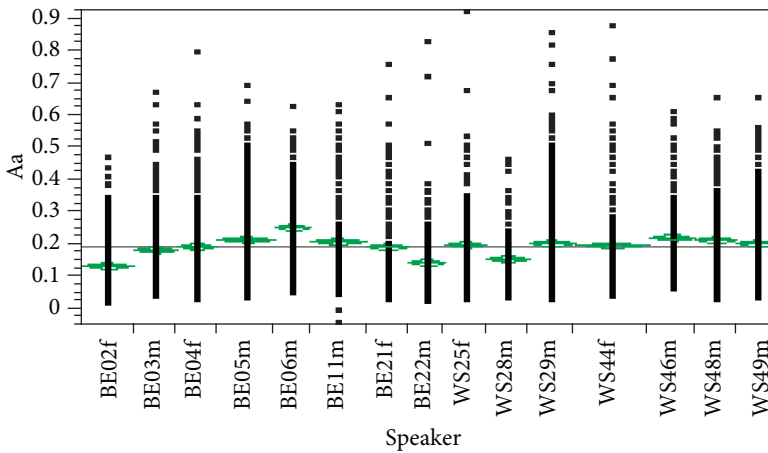


Figure 2. Aa amplitudes. Y-axis: Amplitude, X-axis: speakers. Diamonds mark speaker means and the 95%-confidence interval. The overall mean is marked with the grey horizontal line.

groups are not very distinct, which hints at a control of phrasing at a local and not global level – a finding which supports the results of the earlier study.

When we look at the accent command amplitudes (Figure 2), it is evident that there is not much variation between all 15 speakers.

If we compare the means of the Aa amplitudes of the two groups, however, there are again significant differences between WS and BE (t-test, $p=0.0068$); the WS, again, show higher Aa amplitude values than the BE, which entails that their pitch range on the local level is higher than that of the BE speakers. From these results we deduce that pitch amplitude, i.e., the global and local pitch variation, is crucial when we look at inter-dialectal differences in Swiss German at the intonational level.

4.4 Results IV: WS declarative-final rise

When listening to and analyzing the WS material, we noticed a significant increase in F0 towards the end for a number of declarative sentences, which, according to the talk setting, were categorised as terminating and not continuing phrases. In order to check whether this WS-specific phenomenon could be corroborated with data we investigated the Aa amplitude of the first, medial, penultimate and ultimate syllables in phrases (see Figure 3).

Figure 3 shows the Aa amplitudes of terminating and continuing phrases in the WS group. The accent commands are shown according to their position in the phrase (first, medial, penultimate and ultimate). While, as expected, we find a significant increase in Aa amplitude from the penultimate to the ultimate syllable

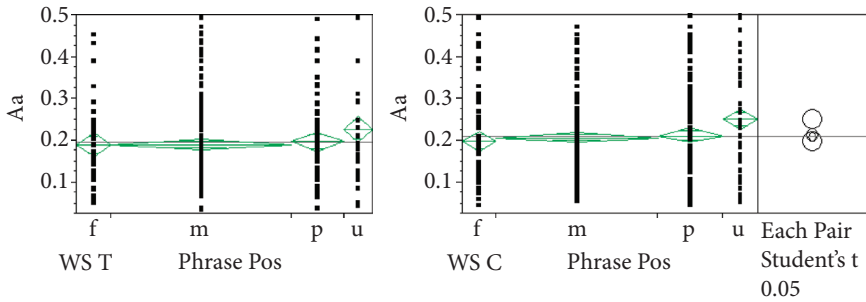


Figure 3. WS Aa amplitudes of terminating (T) and continuing (C) phrases, according to first (f), medial (m), penultimate (p), and ultimate (u) syllable positions in the phrase. Significant differences for student's t-test are found for the factors for which circles do not overlap.

in the continuing phrases (t-test, $p=0.009$), we find a similar, yet not significant pattern for the terminating phrases. The high-rising terminal for continuing phrases is self-evident as speakers want to be cooperative during the interview, which they signal by a rise in F0 towards the end of the phrase and thus encourage the addressee to participate in the conversation. The fact that there exists a similar pattern in terminating phrases for the WS hints at an “uptalk” in a Swiss dialect similar to uptalk in English (cf., e.g., Ladd 1996:123; Warren 2005; Guy et al. 1986).

5. Discussion and conclusion

The present study has shed some light on the intonation of two Swiss German dialects by using a quantitative approach. By means of spontaneous interviews, data was collected and subsequently pre-processed and labelled with PRAAT (2007). The Fujisaki model (1983) was used to analyze intonation contours. Results suggest that the perceptual differences between BE and WS Swiss German are not due to a speculated higher number of accent commands in the WS group, nor to the fact that WS speakers would set more accent commands on unstressed syllables and schwa syllables. We did find, however, that WS speakers stress lexical words more consistently with accent commands and it seems that pitch range accounts for most of the intonational differences between the two dialects. Both the local and global pitch excursions exhibit a wider range in the WS group. The data provided fruitful insight into how BE and WS control their intonation, i.e., more on a local rather than on a global level due to both groups' generally low Ap amplitudes. However, the present analyses need to be refined in several ways.

Firstly, the model will be optimised. As mentioned earlier, we are currently working on the integration of a third component which is geared at modeling

slow rises. Most importantly, the parameters of this slow rise component allow for a more linguistically based approach to the contour decomposition. In a further refinement of the data analysis, the specific points where accent commands start and end with reference to the corresponding syllable need to be taken into consideration. Preliminary analyses have shown that there seem to be significant temporal differences between the two groups with regard to these starting and ending points. This, we assume, makes up for a significant part of the perceptual intonational differences between the two groups. Furthermore, we need to categorise the accent commands, i.e., specify whether the accent commands fall onto a focused element of the phrase, onto a focused element on an unstressed syllable of the phrase, onto merged syllables or onto two subsequent accented syllables. On the paralinguistic level, a categorisation with regard to emotions must be made, where we plan to distinguish between neutral, excited and bored states of the interviewee; this categorisation may provide useful hints for an explanation of specific intonation contours.

Finally, more data will be added to our database, as we are in the process of transcribing, segmenting, and annotating data from the Zurich recordings. Also, recordings in Chur are planned which will allow us to use data from a more elaborate database and thus make more substantiated claims about the findings presented here.

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